Fiscal Year:	FY 2018	**Task Last Updated:**	FY 08/25/2017
PI Name:	Chapes, Stephen K. Ph.D.		
Project Title:	Collection of Immune/Stress-related Tissues from Mice Flown on ISS		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	skcbiol@ksu.edu	**Fax:**	FY 785-532-6653
PI Organization Type:	UNIVERSITY	**Phone:**	785-532-6795
Organization Name:	Kansas State University		
PI Address 1:	1717 Claflin Rd.		
PI Address 2:	Division of Biology		
PI Web Page:			
City:	Manhattan	**State:**	KS
Zip Code:	66506-4901	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	11/30/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
Contact Email:	kevin.y.sato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	Tissue Sharing NOTE: End date changed to 11/30/2018 per F. Hernandez/ARC (Ed., 2/7/17) NOTE: NSSC shows end date of 11/30/2017 (Ed., 8/3/16) NOTE: Extended to 10/31/2016 per A. Chu/ARC and NSSC information (Ed., 10/28/15)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Gridley, Daila Ph.D. (Loma Linda University) Pecaut, Michael Ph.D. (Loma Linda University)		
Grant/Contract No.:	NNX15AB45G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NNX15AB45G was funded in 2014 in response to section B.7 - Special topic - ISS Rodent Tissue Sharing Opportunity announced as part of NRA NNH14ZTT001N. NASA indicated that a total of 20 mice were planned to be flown to the International Space Station (ISS) and 40 mice would serve as ground controls. Four immunization treatment groups were part of the experimental protocol that was designed to assess the in vivo antibody responses during spaceflight. In particular, the antibody response and the B cell immunoglobulin heavy-chain (IgH) repertoire would be determined in the splenic B cells from each of the treatment groups and compared among flight and ground treatments. Our group (Pecaut and Chapes) proposed to collect mouse liver, bone marrow, kidney/adrenal glands, and thymus to accomplish two specific aims. 1. Specific Aim 1: Collect thymus to weigh the tissue to compare relative thymic atrophy among treatment groups and adrenal glands to assess catecholamines. These tissues would have been used to assess short- and long-term stress to the flight animals. 2. Specific Aim 2: Collect liver and bone marrow. These tissues would be used for the preparation of RNA to allow for a comparative analysis of B cell IgH usage between lymphoid compartments. With the loss of astronaut work time assigned to Space Biology and > or = 2 year delay in the transport of mice to the International Space Station, we were approved to modify the aims of NNX15AB45G as follows: 1. Obtain liver tissue/RNA from the mice flown on Rodent Research 1 (RR1) verification flight. Preflight verification tests done as part of NNX15AB45G have shown that usable RNA cannot be obtained from bone marrow from frozen carcasses (using the freezing procedure currently used on ISS). Therefore, we proposed to focus on liver tissue. Liver RNA was available from the liver tissue (10 mice; 5 flight and 5 AEM ground controls). We proposed to do MiSeq next generation sequencing on a group of flight and ground samples. In addition, NASA Gene Lab had done HiSeq next generation sequencing on some of the liver tissue flown on RR1. We proposed to obtain those data. We proposed to determine IgH usage of the liver B cells in the mice (flight vs. ground controls). One of our goals is to determine if there is tissue specificity in B cell IgH usage. 2. An antiorthostatic suspension (AOS) was proposed to skeletally unload mice to simulate the skeletal unloading that occurs during spaceflight (This experiment will be done as part of NNX13AN34G). In this protocol, mice will be subjected to AOS. Two weeks later the mice will be immunized in 1 of 4 ways; PBS, CpG only, tetanus toxoid (TT) only, and TT with CpG. Injections will be intraperitoneal (i.p.). Two weeks after immunization and continued AOS, we will obtain liver tissue from those mice. We will directly assess the changes in IgH usage caused by immunization ± adjuvant (CpG) and will address the impact of skeletal unloading. One of our original goals was to examine the impact of spaceflight on host B cell responsiveness in different tissue compartments. Therefore, this experiment was expected to partially fulfill the goals of Aim 1 of the original grant by using AOS as a spaceflight model. We proposed to compare the impact of skeletal unloading using AOS on host responsiveness to antigen challenge and address the issue of tissue specificity in B cell IgH usage. In addition to the IgH usage studies, we proposed to collect thymus to weigh the tissue to compare relative thymic atrophy among treatment groups and adrenal glands to assess catecholamines. The data obtained from these tissue analyses will be used to assess short- and long-term stress to AOS animals.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Spaceflight offers a unique environment that impacts many physiological systems. This occurs not only in humans but also in rats, mice, and rhesus monkeys. Among these systems, bone and the immune response are connected and are known to be significantly altered by the skeletal unloading of spaceflight. B lymphocytes are one of the cells that develop in the bone marrow. They make antibodies in response to infection or immunization with vaccines. Moreover, the B cells that protect people from disease after receiving immunizations localize to the bone marrow as well as other lymphoid organs. We will obtain valuable information about what changes occur in the B-cell populations that develop in lymphoid organs such as liver and compare the development of these B-cell populations found in other lymphoid compartments. This understanding will tell us if more precautions are necessary on Earth when people are skeletally unloaded. On Earth, skeletal unloading occurs to individuals that are bedridden for long periods of time, wheel-chair bound, or lose the ability to stand for long periods of time. Therefore, the space environment provides an accelerated view of what will happen to individuals that must endure these prolonged periods of skeletal unloading. This experiment will have direct relevance to individuals on Earth.
	Analysis of Rodent Research 1 data. We analyzed gene lab sequencing data that was available from the data base, GLDS-47, GLDS-48. The number of immunoglobulin reads (either heavy or light chain were low; less than 1% of spleen cells and tissue). We first attributed this low yield to short sequencing reads from the HiSeq sequencing method that the GeneLab did and provided in the database. Therefore, we sequenced both splenic (funded by another grant) and liver RNA from the Rodent Research 1 (RR1) validation flight, which was made available by the NASA Ames Research Center (n=3 ground, n=3 flight). RNA was sequenced on the Illumina MiSeq platform (300bp, 2x) at the Kansas State Integrated Genomics Facility. Sequencing data was quality trimmed using CLC genomics workbench and mapped to reference libraries obtained from ImMunoGeneTics (IMGT) information system and the National Center for Biotechnology Information (NCBI) gene database. Datasets were submitted to the IMGT HighV-Quest tool for CDR3 characterization. The analysis of liver vs. spleen reads indicated that liver immunoglobulin gene reads range from 0.26% of spleen immunoglobulin gene reads to 0.91%. These data were presented in more detail in a previous progress report. The number of reads ranged from less than 100 to no greater than 622, depending on the sample. Although useful, publishable data from the spleen can be obtained (preliminary data presented at the 33rd Annual Meeting of the American Society for Gravitational and Space Research, 2016, Poster #120 and presented in the GeneLab workshop by Chapes and Ward) that the liver sequencing did not provide a very useful data set, even when the more comprehensive MiSeq next generation sequencing platform was used to sequence. Of the data obtained, the liver kappa chain V-gene repertoire did correlate some with the usage in the spleen and probably reflects circulating B-cell Ig expression but we decided to switch our focus to the bone marrow (see below). The liver experiment was a component in a paper recently submitted for publication. Although the liver data turned out to be of minimal value for doing tissue comparisons. Other data from the MiSeq NGS data sets will be useful for other data mining attempts and will be made available to the GeneLab by the end of September 2017. Our bioinformatics analyses of these data sets and the description of our bioinformatics workflow was published in Gravitational Space Research, 5(1):2-23, 2017. This work was also highlighted on the cover of GSR (http://gravitationalandspacebiology.org/).

Task Progress:	An antiorthostatic suspension (AOS) was proposed to skeletally unload mice to simulate the skeletal unloading that occurs during space flight (This experiment was done in conjunction with Dr. Pecaut, NNX13AN34G). Mice were subjected to AOS. Two weeks after the start of unloading, mice were immunized (i.p.) in 1 of 4 ways; PBS, CpG only, tetanus toxoid (TT) only, and TT with CpG. Two weeks after immunization and continued AOS, we obtained liver tissue from those mice. We proposed to assess the changes in immunoglobulin gene usage caused by immunization ± adjuvant (CpG) and wanted to address the impact of skeletal unloading. One of our original goals was to examine the impact of spaceflight on host B cell responsiveness in different tissue compartments. Spleen tissue was to be analyzed. Therefore, this experiment (NNX15AB45G) was proposed to attempt to see how using AOS as a space flight model would affect host responsiveness to antigen challenge and would have allowed a comparison of spleen and liver tissue (tissue specificity). Progress Completion of Antiorthostatic Suspension (AOS) Experiment. The AOS was done to skeletally unload mice to simulate the skeletal unloading and stress that occurs during space flight (Chapes, Mastro et al. 1993) (Experiment 1 took place in March 2016 [5 mice per treatment group] ; Experiment 2 took place in May 2016 [5 mice per treatment group]). In this protocol, female, 8-week-old mice were subjected to AOS. Two weeks later the mice were immunized in 1 of 4 ways; PBS (buffer control), CpG only (adjuvant), tetanus toxoid (TT; antigen) only, and TT with CpG. Injections were done by intraperitoneal (i.p.) route. Two weeks after immunization and continued AOS (28 day total unloading), we obtained both spleen and liver tissue from those mice. This protocol was followed so that when mice are flown to ISS in the future, we will have the AOS as a comparison group with parallel variables. Some of these data were recently presented at the 33rd Annual Meeting of the American Society for Gravitational and Space Research (Poster #67). Our original aims were to determine if there was tissue specificity in B cell IgH usage along with assessing the impact of space flight on immunoglobulin gene usage. Bone marrow is an important, essential immune compartment for the storage of B cell plasma and memory cells (Erickson, Lin et al. 2003; Cariappa, Chase et al. 2007). Therefore, we were approved to shift our efforts to examine the bone marrow that was collected from an AOS mouse experiment that was done as part of another project NNX13AN34G. Bone marrow RNA has been isolated from AOS samples and RNA has been submitted for sequencing at the Integrated Genomics Facility at Kansas State University. These sequences are now being completed and analyzed and we were able to identify Ig sequences in the bone marrow suitable for analysis. 32-bone marrow samples will be sequenced and analyzed in the coming months. References Chapes, S. K., A. Mastro, G. Sonnenfeld and W. Berry (1993). "Antiorthostatic suspension as a model for the effects of spaceflight on the immune system." J. Leukoc. Biol. 54: 227-235. Erickson, L. D., L.-L. Lin, B. Duan, L. Morel and R. J. Noelle (2003). "A genetic lesion that arrests plasma cell homing to the bone marrow." PNAS: 2131686100. Cariappa, A., C. Chase, H. Liu, P. Russell and S. Pillai (2007). "Naive recirculating B cells mature simultaneously in the spleen and bone marrow." Blood 109(6): 2339-2345.
Bibliography Type:	Description: (Last Updated: 09/15/2021)
Abstracts for Journals and Proceedings	Binaco PJ, Ayala S, Howe D, Pecaut MJ, Nishiyama NC, Mao XM, Rodriguez D, Kwok A, Bateman TA, Chapes SK, Willey JS, Lau AG. "Effects of Low Dose Radiation and Tetanus Toxoid on the Strength of Bone." Presented at the 2016 Biomedical Engineering Society (BMES) Annual Meeting, Minneapolis, MN, October 5-8, 2016. 2016 Biomedical Engineering Society (BMES) Annual Meeting, Minneapolis, MN, October 5-8, 2016. , Oct-2016
Abstracts for Journals and Proceedings	Ward C, Rettig TA, Pecaut MJ, Chapes SK. "Using GeneLab data to assess immunoglobulin gene segment usage." Presented at the 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Abstracts for Journals and Proceedings	Hlavacek S, Bye B, Ward C, Rettig T, Chapes SK. "Assessing the impact of space flight on antibody gene segment pairing in C57BL/6 mice." Presented at the 15th Annual K-INBRE Symposium, Manhattan, KS, January 14-15, 2017. 15th Annual K-INBRE Symposium, Manhattan, KS, January 14-15, 2017. , Jan-2017
Abstracts for Journals and Proceedings	Chapes SK, Rettig TA, Ward C, Nishiyama NC, Bye B, Hlavacek S, Pecaut MJ. "Assessment of mouse immunoglobulin gene-segment usage by mice in space flight and space flight models." Presented at the 2017 International Space Station (ISS) Research and Development Conference, Washington, DC, July 17-20, 2017. 2017 International Space Station (ISS) Research and Development Conference, Washington, DC, July 17-20, 2017. , Jul-2017
Articles in Peer-reviewed Journals	Shi Z, Chapes SK, Ben-Arieh D, Wu CH. "An agent-based model of a hepatic inflammatory response to Salmonella: A computational study under a large set of experimental data." PLoS One. 2016 Aug 24;11(8):e0161131. eCollection 2016. https://doi.org/10.1371/journal.pone.0161131 ; PubMed PMID: 27556404; PubMed Central PMCID: PMC4996536 , Aug-2016
Articles in Peer-reviewed Journals	Kurti SP, Rosenkranz SK, Chapes SK, Teeman CS, Cull BJ, Emerson SR, Levitt MH, Smith JR, Harms CA. "Does chronic physical activity level modify the airway inflammatory response to an acute bout of exercise in the post-prandial period?" Appl Physiol Nutr Metab. 2017 Feb;42(2):173-80. Epub 2016 Oct 18. https://doi.org/10.1139/apnm-2016-0335 ; PubMed PMID: 28121185; PubMed Central PMCID: PMC5357083 , Feb-2017
Articles in Peer-reviewed Journals	Rettig TA, Ward C, Pecaut MJ, Chapes SK. "Validation of methods to assess the immunoglobulin gene repertoire in tissues obtained from mice on the International Space Station." Gravit Space Res. 2017 Jul;5(1):2-23. http://gravitationalandspacebiology.org/index.php/journal/article/view/750 , Jul-2017

Journal/Magazine covers	Rettig TA, Ward C, Pecaut MJ, Chapes SK. "Cover in journal Gravitational Space Research for article, 'Validation of methods to assess the immunoglobulin gene repertoire in tissues obtained from mice on the International Space Station.' " Gravit Space Res. 2017 Jul;5(1):2-23. http://gravitationalandspacebiology.org/index.php/journal , Jul-2017